Therefore the number of numbers formed in $5 \times 4 \times 3 \times 2 = 120$ ways.

$\therefore$ 120 four digit numbers can be formed.

Factorial:

The consecutive product of first n natural numbers is known as factorial n and is denoted as $n!$ or $\underline{|n}$

That is $n! = n \times (n-1) \times \ldots \times 3 \times 2 \times 1$

$3! = 3 \times 2 \times 1$

$4! = 4 \times 3 \times 2 \times 1$

$5! = 5 \times 4 \times 3 \times 2 \times 1$

$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$

Also $6! = 6 \times (5 \times 4 \times 3 \times 2 \times 1) = 6 \times (5!)$

This can be algebraically expressed as $n! = n\,(n-1)!$

Note that $1! = 1$ and $0! = 1$.

7.2 Permutations

Permutation means arrangement of things in different ways. Let us take 3 things A, B, C for an arrangement. Out of these three things two at a time, we can arrange them in the following manner.

AB	AC	BC
BA	CA	CB

Here we find 6 arrangements. In these arrangements, order of arrangement is considered. Note that the arrangement AB and the arrangement BA are different.

The number of arrangements of the above is given as the number of permutation of 3 things taken 2 at a time which gives the value 6.

This is written symbolically $3P_2 = 6$

Thus the number of arrangements that can be formed out of n things taken r at a time is known as the number of permutation of n things taken r at a time and is denoted as nP_r or $P(n, r)$

We write nP_r as $nP_r = n\,(n-1)\,(n-2)\,\ldots\,[n-(r-1)]$